

SKKU 2024 Challenge - Quantum State Tomography

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November 7, 2024

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Problem 1: Quantum States with Few Qubits

The first class of states that we will consider are generic pure quantum states (i.e. there is no restriction on the quantum circuit which created the state), on a small number of qubits. In general, a generic quantum state will require an exponential number of measurements to reconstruct. However, when the number of qubits is small, we can simply pay this exponential cost.

Part I

In this first problem, we have a single qubit quantum state

$$|\psi\rangle = a|0\rangle + be^{i\theta}|1\rangle, \quad (1)$$

where we can treat the unknown parameters a, b and θ as real positive numbers with $|a|^2 + |b|^2 = 1$ and $\theta \in [0, 2\pi)$.

a) Explain how one can determine the values of a, b and θ by measuring the quantum state in different bases.

Start with the given single qubit pure state:

$$|\psi\rangle = a|0\rangle + be^{i\theta}|1\rangle \quad (2)$$

where a, b, θ are real parameter subject to $a^2 + b^2 = 1$ and $0 \leq \theta < 2\pi$. In order to determine these three parameter, it requires at least three measurement. Our measurement is performed via POVM. There are detectors whose number is equal to the number of qubits. In this scenerio, we perform z -basis projection measurement. Therefore, the POVM becomes $\{|0\rangle\langle 0|, |1\rangle\langle 1|\}$, each corresponds to the presense and absence of the signal. Let $\hat{m}_0, \hat{m}_1$ denote the estimated probability of occuring signals. We can

add additional operation U before the measurement and after the computational stage(here, state preparation), we can perform measurement another POVM basis, $\{U_{\text{add}}^\dagger |0\rangle\langle 0| U_{\text{add}}, U_{\text{add}}^\dagger |1\rangle\langle 1| U_{\text{add}}\}$. As we should run several circuits, let U_{add^i} denote the additional circuit to i -th circuit.

In first measurement, to determine a, b , we can directly perform measurement without any unitary operation. That is $U_{\text{add}^1} = I$, here. Then, we see:

$$\hat{m}_0 \simeq |\langle \psi | 0 \rangle|^2 = a^2 \quad (3)$$

$$\hat{m}_1 \simeq |\langle \psi | 1 \rangle|^2 = b^2 \quad (4)$$

Therefore, we can estimate,

$$a = \sqrt{\hat{m}_0} \quad (5)$$

$$b = \sqrt{\hat{m}_1} \quad (6)$$

On the other hand, we need two additional measurement. Here let $U_{\text{add}^2} = H$, where H is a hadamard gate.

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \quad (7)$$

Then, our POVM becomes $\{|+\rangle\langle +|, |-\rangle\langle -|\}$ where $|+\rangle, |-\rangle$ are the eigenstate of Pauli-X matrix whose eigenvalue is 1, -1 for each. Here, let $\hat{m}_+, \hat{m}_-$ be the estimated probability of occuring signals.

$$\hat{m}_+ \simeq |\langle \psi | + \rangle|^2 = \left| \frac{a + be^{i\theta}}{2} \right|^2 = \frac{1 + 2ab \cos \theta}{2} \quad (8)$$

$$\hat{m}_- \simeq |\langle \psi | - \rangle|^2 = \left| \frac{a - be^{i\theta}}{2} \right|^2 = \frac{1 - 2ab \cos \theta}{2} \quad (9)$$

Or,

$$\cos \theta = \frac{\hat{m}_+ - \hat{m}_-}{2ab} \quad (10)$$

Here, it is not enough yet. As there is a degree of freedom choosing a sign of $\sin \theta$. Finally, let $U_{\text{add}^3} = SH$ where S is S gate.

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}. \quad (11)$$

Then our POVM becomes $\{|i\rangle\langle i|, |-i\rangle\langle -i|\}$ where $|i\rangle, |-i\rangle$ are the eigenstate of Pauli-Y matrix whose eigenvalue is 1, -1 for each. Here, let $\hat{m}_i, \hat{m}_{-i}$ be the estimated probability of occuring signals.

$$\hat{m}_i \simeq |\langle \psi | i \rangle|^2 = \left| \frac{a + be^{i(\theta + \frac{\pi}{2})}}{2} \right|^2 = \frac{1 - 2ab \sin \theta}{2} \quad (12)$$

$$\hat{m}_{-i} \simeq |\langle \psi | -i \rangle|^2 = \left| \frac{a - be^{i\theta + \frac{\pi}{2}}}{2} \right|^2 = \frac{1 + 2ab \sin \theta}{2} \quad (13)$$

Or,

$$\sin \theta = -\frac{\hat{m}_i + \hat{m}_{-i}}{2ab}. \quad (14)$$

Therefore, we can determine a, b and θ .

$$a = \sqrt{\hat{m}_0} \quad (15)$$

$$b = \sqrt{\hat{m}_1} \quad (16)$$

$$\theta = \arcsin \frac{-\hat{m}_i + \hat{m}_{-i}}{2ab} = \arccos \frac{\hat{m}_+ - \hat{m}_-}{2ab} \quad (17)$$

b) Write a program in Qiskit which generates this unknown quantum state.

To generate a random single qubit quantum state we use a circuit U_{hidden} , so that $|\psi\rangle = U_{\text{hidden}}|0\rangle$. I used the following provided code, to generate random state.

```
def create_1q_target_circuit(qc_data):
    qc = qiskit.QuantumCircuit(1)
    qc.ry(qc_data[0],0)
    qc.rz(qc_data[1],0)
    return qc
```

As discussed in a), we can determine the state in Eq. 2 via three measurements. To reconstruct the statevector, I implemented the following code.

```
def SingleQubitQST(U_hidden, num_shots:int=100):
    hidden_circuit = U_hidden.copy()

    qc1 = qiskit.QuantumCircuit(1)
    circ1 = hidden_circuit.compose(qc1)
    circ1.measure_all()
    #print(circ1.draw())

    qc2 = qiskit.QuantumCircuit(1)
    qc2.h(0)
    circ2 = hidden_circuit.compose(qc2)
    circ2.measure_all()
    #print(circ2.draw())

    qc3 = qiskit.QuantumCircuit(1)
    qc3.s(0)
```

```

qc3.h(0)
circ3 = hidden_circuit.compose(qc3)
circ3.measure_all()
#print(circ3.draw())

result1 = MeasurementResultHandler(backend
    .run(circ1, shots=num_shots)
    .result().data()["counts"], 1, num_shots)
result2 = MeasurementResultHandler(backend
    .run(circ2, shots=num_shots)
    .result().data()["counts"], 1, num_shots)
result3 = MeasurementResultHandler(backend
    .run(circ3, shots=num_shots)
    .result().data()["counts"], 1, num_shots)

m0 = result1.get("0x0"); m1 = result1.get("0x1")
if m0 == 1:
    return np.array([1,0], dtype=np.complex64)
elif m1 == 1:
    return np.array([0,1], dtype=np.complex64)

phi = np.arctan2(np.sqrt(m1), np.sqrt(m0))
phi = phi if phi > 0 else phi + 2*np.pi
statevector = np.array([np.cos(phi),
                        np.sin(phi)],
                        dtype=np.complex64)

m_p = result2.get("0x0"); m_m = result2.get("0x1")
cos = (m_p - m_m)/(2*np.cos(phi)*np.sin(phi))

m_pi = result3.get("0x0"); m_mi = result3.get("0x1")
sin = (-m_pi + m_mi)/(2*np.cos(phi)*np.sin(phi))

temp = np.array([cos, sin])

cos = cos / np.linalg.norm(temp)
sin = sin / np.linalg.norm(temp)
print(circ1.draw())
theta = np.arctan2(sin, cos)
theta = theta if theta > 0 else theta + 2*np.pi

statevector[1] *= np.exp(1j*theta)
circuit_parameter = np.array([2*phi, theta])

```

```
return statevector, circuit_parameter
```

Then, we determine the circuit parameters from the reconstructed state vector. Using ZYZ decomposition, we can represent all the unitary operation on a single qubit. Here, for state preparation, we only need RY and RZ gate[2, 3, 1]. The state would be

$$|\psi\rangle = R_z(\beta)R_y(\alpha) = \cos\left(\frac{\alpha}{2}\right)|0\rangle + e^{i\beta}\sin\left(\frac{\alpha}{2}\right)|1\rangle. \quad (18)$$

without global phase. Comparing this with the Eq. 2, we can determine circuit parameter α, β .

$$\alpha = 2 \arccos a = 2 \arcsin b \quad (19)$$

$$\beta = \theta \quad (20)$$

The following function performs determining circuit parameters and generating U_{gen} .

```
def SingleQubitCircuit(statevector):
    cos = np.linalg.norm(statevector[0])
    sin = np.linalg.norm(statevector[1])
    phi = np.arctan2(sin,cos)
    theta = (np.log(statevector[1]/
                    np.linalg.norm(statevector[1]))*-1j).real if statevector[1] != 0 else 0

    U_gen = qiskit.QuantumCircuit(1)
    U_gen.ry(2*phi,0)
    U_gen.rz(theta,0)

    return U_gen
```

To evaluate this reconstruction scheme, the following provided code is used.

```
backend2 = Aer.get_backend("statevector_simulator")
def give_score(U_hidden, U_gen):
    qc = U_hidden.compose(U_gen.inverse())
    result = backend2.run(qc).result()
    score = result.get_statevector()[0]
    return np.abs(score)
```

Part II

We will now increase the complexity by reconstructing a generic two-qubit quantum state.

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle \quad (21)$$

where $|a|^2 + |b|^2 + |c|^2 + |d|^2 = 1$.

a) First, consider the case where all amplitudes are real numbers. Explain how you can measure both the magnitude and sign of the unknown amplitudes, measuring in as few different bases as possible.

If the parameters are subjected to real numbers, it seems sufficient to identify all the parameters a, b, c and d with running three circuits. In first circuit, measurement amplitudes. That is, $U_{\text{add}}^0 = I_4$.

$$\hat{m}_{00} \simeq |\langle\psi|00\rangle|^2 = a^2 \quad (22)$$

$$\hat{m}_{01} \simeq |\langle\psi|01\rangle|^2 = b^2 \quad (23)$$

$$\hat{m}_{10} \simeq |\langle\psi|10\rangle|^2 = c^2 \quad (24)$$

$$\hat{m}_{11} \simeq |\langle\psi|11\rangle|^2 = d^2 \quad (25)$$

Then,

$$a = \sqrt{m_{00}} \quad (26)$$

$$b = \sqrt{m_{01}} \quad (27)$$

$$c = \sqrt{m_{10}} \quad (28)$$

$$d = \sqrt{m_{11}} \quad (29)$$

$$(30)$$

Next, run and measure the circuit with Hadamard gate on first qubit $U_{\text{add}}^1 = I \otimes H$.

$$\hat{m}_{0+} \simeq |\langle\psi|0+\rangle|^2 = \frac{a+b}{2} \quad (31)$$

$$\hat{m}_{0-} \simeq |\langle\psi|0-\rangle|^2 = \frac{a-b}{2} \quad (32)$$

$$\hat{m}_{1+} \simeq |\langle\psi|1+\rangle|^2 = \frac{c+d}{2} \quad (33)$$

$$\hat{m}_{1-} \simeq |\langle\psi|1-\rangle|^2 = \frac{c-d}{2} \quad (34)$$

Finally, run and measure the circuit with Hadamard gate on first qubit $U_{\text{add}}^2 = H \otimes I$.

$$\hat{m}_{+0} \simeq |\langle \psi | +0 \rangle|^2 = \frac{a+c}{2} \quad (35)$$

$$\hat{m}_{+1} \simeq |\langle \psi | +1 \rangle|^2 = \frac{a-c}{2} \quad (36)$$

$$\hat{m}_{-0} \simeq |\langle \psi | -0 \rangle|^2 = \frac{b+d}{2} \quad (37)$$

$$\hat{m}_{-1} \simeq |\langle \psi | -1 \rangle|^2 = \frac{b-d}{2} \quad (38)$$

Now, compare the measured result. a is fixed to positive, and if $\hat{m}_{0+} > \hat{m}_{0-}$, then b is also positive, otherwise b is negative. If $\hat{m}_{+0} > \hat{m}_{-0}$, then c is also positive, otherwise c is negative. After determining, b and c , we can determine the sign of d . If $\hat{m}_{1+} > \hat{m}_{1-}$ ($\hat{m}_{-0} > \hat{m}_{-1}$), then d will have the same sign with $b(c)$, otherwise will have opposite sign.

b) Now, explain how you would adapt this measurement protocol if the amplitudes a, b, c, d are complex numbers. Note, that we can only reconstruct the wavefunction up to a global phase, so that we can always consider a to be a positive real number.

The two-qubit pure state can be written as:

$$|\psi\rangle = a|00\rangle + be^{i\theta_b}|01\rangle + ce^{i\theta_c}|10\rangle + de^{i\theta_d}|11\rangle \quad (39)$$

where a, b, c and d are real positive numbers $\theta_b, \theta_c, \theta_d$ are real numbers subject to $0 \leq \theta_b, \theta_c, \theta_d < 2\pi$ without loss of generality. First, we run circuit without any additional operation, $U_{\text{add}}^0 = I_4$. Then our POVM bases are $\{|00\rangle\langle 00|, |01\rangle\langle 01|, |10\rangle\langle 10|, |11\rangle\langle 11|\}$, which directly reproduce a^2, b^2, c^2 and d^2 . That is,

$$\hat{m}_{00} \simeq |\langle \psi | 00 \rangle|^2 = a^2 \quad (40)$$

$$\hat{m}_{01} \simeq |\langle \psi | 01 \rangle|^2 = b^2 \quad (41)$$

$$\hat{m}_{10} \simeq |\langle \psi | 10 \rangle|^2 = c^2 \quad (42)$$

$$\hat{m}_{11} \simeq |\langle \psi | 11 \rangle|^2 = d^2 \quad (43)$$

Next, using Hadamard gate on first qubit $U_{\text{add}}^1 = I_2 \otimes H$, we have:

$$\hat{m}_{0+} \simeq |\langle \psi | 0+ \rangle|^2 = \left| \frac{a + be^{i\theta_b}}{\sqrt{2}} \right|^2 \quad (44)$$

$$\hat{m}_{0-} \simeq |\langle \psi | 0- \rangle|^2 = \left| \frac{a - be^{i\theta_b}}{\sqrt{2}} \right|^2 \quad (45)$$

$$\hat{m}_{1+} \simeq |\langle \psi | 1+ \rangle|^2 = \left| \frac{ce^{i\theta_c} + de^{i\theta_d}}{\sqrt{2}} \right|^2 \quad (46)$$

$$\hat{m}_{1-} \simeq |\langle \psi | 1- \rangle|^2 = \left| \frac{ce^{i\theta_c} - de^{i\theta_d}}{\sqrt{2}} \right|^2 \quad (47)$$

Then, we see:

$$\cos \theta_b = \frac{\hat{m}_{0+} - \hat{m}_{0-}}{2ab} \quad (48)$$

$$\cos(\theta_c - \theta_d) = \frac{\hat{m}_{1+} - \hat{m}_{1-}}{2cd} \quad (49)$$

$$(50)$$

With Hadamard gate and S gate on first qubit, $U_{\text{add}}^2 = I_2 \otimes SH$, we see:

$$\hat{m}_{0i} \simeq |\langle \psi | 0i \rangle|^2 = \left| \frac{a + ce^{i\theta_c + \frac{\pi}{2}}}{\sqrt{2}} \right|^2 \quad (51)$$

$$\hat{m}_{0-i} \simeq |\langle \psi | 0-i \rangle|^2 = \left| \frac{be^{i\theta_b} + de^{i\theta_d + \frac{\pi}{2}}}{\sqrt{2}} \right|^2 \quad (52)$$

$$\hat{m}_{1i} \simeq |\langle \psi | 1i \rangle|^2 = \left| \frac{a - ce^{i\theta_c - \frac{\pi}{2}}}{\sqrt{2}} \right|^2 \quad (53)$$

$$\hat{m}_{1-i} \simeq |\langle \psi | 1-i \rangle|^2 = \left| \frac{be^{i\theta_b} - de^{i\theta_d - \frac{\pi}{2}}}{\sqrt{2}} \right|^2 \quad (54)$$

Then,

$$\sin \theta_b = \frac{-\hat{m}_{0i} + \hat{m}_{0-i}}{2ab} \quad (55)$$

$$\sin(\theta_c - \theta_d) = \frac{\hat{m}_{1i} - \hat{m}_{1-i}}{2cd} \quad (56)$$

$$(57)$$

Also, using Hadamard gate on second qubit, $U_{\text{add}}^3 = H \otimes I_2$, we have:

$$\hat{m}_{+0} \simeq |\langle \psi | +0 \rangle|^2 = \left| \frac{a + ce^{i\theta_c}}{\sqrt{2}} \right|^2 \quad (58)$$

$$\hat{m}_{+1} \simeq |\langle \psi | +1 \rangle|^2 = \left| \frac{be^{i\theta_b} + de^{i\theta_d}}{\sqrt{2}} \right|^2 \quad (59)$$

$$\hat{m}_{-0} \simeq |\langle \psi | -0 \rangle|^2 = \left| \frac{a - ce^{i\theta_c}}{\sqrt{2}} \right|^2 \quad (60)$$

$$\hat{m}_{-1} \simeq |\langle \psi | -1 \rangle|^2 = \left| \frac{be^{i\theta_b} - de^{i\theta_d}}{\sqrt{2}} \right|^2 \quad (61)$$

Then, we see:

$$\cos \theta_c = \frac{\hat{m}_{+0} - \hat{m}_{-0}}{2ac} \quad (62)$$

$$\cos(\theta_b - \theta_d) = \frac{\hat{m}_{+1} - \hat{m}_{-1}}{2bd} \quad (63)$$

$$(64)$$

With Hadamard gate and S gate on second qubit, $U_{\text{add}}^4 = SH \otimes I_2$, we see:

$$\hat{m}_{i0} \simeq |\langle \psi | i0 \rangle|^2 = \left| \frac{a + ce^{i\theta_c + \frac{\pi}{2}}}{\sqrt{2}} \right|^2 \quad (65)$$

$$\hat{m}_{i1} \simeq |\langle \psi | i1 \rangle|^2 = \left| \frac{be^{i\theta_b} + de^{i\theta_d + \frac{\pi}{2}}}{\sqrt{2}} \right|^2 \quad (66)$$

$$\hat{m}_{-i0} \simeq |\langle \psi | -i0 \rangle|^2 = \left| \frac{a - ce^{i\theta_c + \frac{\pi}{2}}}{\sqrt{2}} \right|^2 \quad (67)$$

$$\hat{m}_{-i1} \simeq |\langle \psi | -i1 \rangle|^2 = \left| \frac{be^{i\theta_b} - de^{i\theta_d + \frac{\pi}{2}}}{\sqrt{2}} \right|^2 \quad (68)$$

Then,

$$\sin \theta_c = \frac{-\hat{m}_{i0} + \hat{m}_{-i0}}{2ac} \quad (69)$$

$$\sin(\theta_b - \theta_d) = \frac{\hat{m}_{i1} - \hat{m}_{-i1}}{2bd} \quad (70)$$

$$(71)$$

Now we can determine $\theta_b, \theta_c, \theta_d$.

Error Handling However, a problem arises when some of a, b, c, d are zero.

- if $a = 0$, Eq. 48, Eq. 55, Eq. 62, Eq. 69 are not available.
- if $b = 0$, Eq. 48, Eq. 55, Eq. 63, Eq. 70 are not available.
- if $c = 0$, Eq. 49, Eq. 56, Eq. 62, Eq. 69 are not available.
- if $d = 0$, Eq. 49, Eq. 56, Eq. 63, Eq. 70 are not available.

When $x(x \in \{b, c, d\})$ is zero, after letting $\theta_x = 0$, we can find the other phase parameter θ by using the other available equations. When a is zero, we can make $\theta_b = 0$. The other cases can be covered by this solution, except two cases: $a = d = 0$ and $b = c = 0$. There is no way to measure phase difference in such case. To resolve this issue, after running the first circuit and determining whether error handling is required, I add additional CNOT gate, which q_0 is control qubit and q_1 is target qubit, to the other circuits. Therefore, we have,

$$U_{\text{add}}^1 = (I_2 \otimes H)CNOT_{0,1} \quad (72)$$

$$U_{\text{add}}^2 = (I_2 \otimes SH)CNOT_{0,1} \quad (73)$$

$$U_{\text{add}}^3 = (H \otimes I_2)CNOT_{0,1} \quad (74)$$

$$U_{\text{add}}^4 = (SH \otimes I_2)CNOT_{0,1} \quad (75)$$

$$(76)$$

Then, the coefficients of $|01\rangle$ and $|11\rangle$ are flipped, then we can treat it as if $a = b = 0$ and $c = d = 0$ case. After reconstructing state vector, we can just change the coefficients of $|01\rangle$ and $|11\rangle$.

c) Now write a program in Qiskit which will generate this unknown quantum state. Use the same format as for part I. To generate random state the following provided code is used:

```
def create_2q_target_circuit(qc_data):
    qc = qiskit.QuantumCircuit(2)
    for i in range(2):
        qc.ry(qc_data[2*i+1], i)
        qc.rz(qc_data[2*i+1], i)
    qc.cz(0, 1)
    qc.rz(qc_data[4], 1)
    qc.cx(0, 1)
    for i in range(2):
        qc.ry(qc_data[2*i+5], i)
```

```
qc.rz(qc_data[2*i+5],i)
return qc
```

As discussed in b), we can reconstruct state vector with 5 measurements. I implemented the following code to reconstruct state vector.

```
def TwoQubitQST(U_hidden, num_shots:int=100):
    hidden_circuit = U_hidden.copy()
```

Now we should determine circuit parameters and ansatz. It is known that one CNOT gate is enough to prepare two qubit state. So I used the following circuit ansatz.

For bipartite pure state $|\psi\rangle$, there exist orthonormal state $|i\rangle$ the state for the first party, $|j\rangle$ the state for the other party such that

$$|\psi\rangle = \sum_i \lambda_i |i_1\rangle \otimes |i_2\rangle \quad (77)$$

where λ_i is non-negative real numbers with $\sum_i \lambda_i^2 = 1$. It is known to schmidt decomposition. By the singular value decomposition, $\lambda = UDV$, where D is a diagonal matrix with non-negative elements, and U, V are unitary matrices. Then,

$$|\psi\rangle = \sum_{i,j,k} U_{ji} D_{ii} V_{ik} |j\rangle |k\rangle. \quad (78)$$

Defining $|i_1\rangle = \sum_j U_{ji} |j\rangle$, $|i_2\rangle = \sum_j V_{ik} |k\rangle$ and $\lambda_i = d_{ii}$, we have the same form of Eq. 77. But here, for convenience in circuit construction, make $\langle 0|i_1\rangle$ and $\langle 0|i_2\rangle$ be real number. It can be done by extracting global phase out, then λ_i is no longer real number. Also, we can extract global phase out, $\lambda'_i = \lambda_i \frac{|\lambda_0|}{\lambda_0}$ so that λ'_0 to be real number.

Here is a strategy to make circuit ansatz and determine circuit parameter. From the initial state, $|\psi\rangle = |00\rangle$, by using local unitary gate on the first qubit, we have:

$$|\psi\rangle \rightarrow \lambda'_0 |00\rangle + \lambda'_1 |01\rangle \quad (79)$$

Then, apply CNOT gate.

$$|\psi\rangle \rightarrow \lambda'_0 |00\rangle + \lambda'_1 |11\rangle \quad (80)$$

Then, applying local unitary operation U_0, U_1 on each qubit, such that

$$U_0 |0\rangle \rightarrow |0\rangle \quad U_0 |1\rangle \rightarrow |0\rangle \quad (81)$$

$$U_1 |0\rangle \rightarrow |0\rangle \quad U_1 |1\rangle \rightarrow |0\rangle \quad (82)$$

Finally, the state becomes our reconstructed state in schmidt decomposition form in Eq. 77.

Now local unitary matters. First, we need to make unitary, $|0_1\rangle \rightarrow \lambda'_0 |0\rangle + \lambda'_1 |1\rangle$. This can be done YZ decomposition, as done in Part I.

$$R_z(\beta_1)R_y(\alpha_1)|0\rangle = \cos \frac{\alpha}{2} |0\rangle + e^{i\beta} \sin \frac{\alpha}{2} |1\rangle \quad (83)$$

Then, we have

$$\alpha_1 = 2 \arccos \lambda'_0 = 2 \arcsin |\lambda'_1| \quad (84)$$

$$\beta_1 = -i \ln \frac{\lambda_1}{|\lambda_1|}. \quad (85)$$

Next, two local unitary U_1, U_2 should be implemented. However, as $|0\rangle$ and $|1\rangle$ orthonormal basis, it can also implemented as if we are doing state preparation, by using Ry, Rz gate. We made $\langle 0|i_1\rangle$ and $\langle 0|i_2\rangle$ be real number, so

$$\alpha_2 = 2 \arccos \langle 0|i_1\rangle \quad (86)$$

$$\beta_2 = -i \ln \frac{\langle 1|i_1\rangle}{|\langle 1|i_1\rangle|} \quad (87)$$

$$\alpha_3 = 2 \arccos \langle 0|i_2\rangle \quad (88)$$

$$\beta_3 = -i \ln \frac{\langle 1|i_2\rangle}{|\langle 1|i_2\rangle|} \quad (89)$$

Therefore, my algorithm procedure is as following:

Phase 1. Perform Schmidt decomposition on reconstructed vector. Numpy has schmidt decomposition method. The column vector in U becomes the basis $|i_1\rangle$ and row vector in $V^\dagger$ are the basis $|i_2\rangle$.

Phase 2. Extract global phase for all schmidt decomposed basis so that the first element of the basis vector to be real. It can done by dividing global phase to state vector and multiplying it to λ_i . And extract global phase one more time, so that λ'_0 be real.

Phase 3. Calculate circuit parameters, by using Eq. 84 and Eq. 89.

Phase 4. Construct circuit.

Exception A problem also arises here: the non-zero schmidt coefficients may not be two. This is the case where the given state is actually separable state, does not require any two-qubit gate in state preparation. Also, schmidt coefficient must be 1, we just implemente two local unitary operation to each qubit as done in Eq. 81 and Eq. 82. The process of extracting of circuit parameter is same with from Eq. 86 to Eq. 89.

Part III

Describe how you could extend the protocol developed in part's I) and II) to a general n qubit quantum state. How many different measurements would you need to apply?

The reconstruction of quantum state constitutes two parts: state reconstruction and circuit construction. An arbitrary n -qubit pure state can be written as:

$$|\psi\rangle = \sum_{i=0}^{2^n-1} a_i e^{i\theta_i} |i\rangle \quad (90)$$

where i denotes bit string, a_i are real positive number under the constraint $\sum_i |a_i|^2 = 1$, $0 \leq \theta_i < 2\pi$ and $\theta_0 = 0$ without loss of generality. First, state reconstruction can be done two step, extract amplitudes and measure phase. The amplitudes a_i can be easily extracted by measuring on a basis $\{|0\rangle\langle 0|, |1\rangle\langle 1|, \dots, |2^n-1\rangle\langle 2^n-1|\}$, that is, without any additional operation before measurement and after state preparation. The phase difference between $|i\rangle$ and $|j\rangle$, where hamming distance between i and j is 1, can be conducted by using additional unitary operation on i -th qubit, $U = H_i$ and $U = S_i H_i$. Then we can determine phase difference $\theta_i - \theta_j$ as done in Part I and Part II. In order to determine whole phase difference, such measurement, with additional unitary operation on i -th qubit, $U = H_i$ and $U = S_i H_i$, should be done for all qubits. Therefore, we need $2n + 1$ circuits to perform QST on n -qubit pure state.

Once state vector is reconstructed, we can determine circuit ansatz and parameters. Up to three qubits, there are well known ansatz to prepare arbitrary states. One and two qubit cases are discussed above Part I and Part II. For three qubits, it is well known that three CNOT gate is sufficient for preparation of arbitrary three-qubit pure state. Or this reference can be another example[4]. For $n \geq 4$, here are strategy[2, 5].

Phase 1 If n is an even number, $n = 2k$, then, by Schmidt decomposition, we have:

$$|\psi\rangle = \sum_{i=0}^{2^k-1} \alpha_i |i_1\rangle |i_2\rangle \quad (91)$$

Phase 2 In a similar way to the Part II, apply local unitary gate to initial state, $ket\psi = ket0^{\otimes n}$, such that

$$ket\psi \rightarrow \left(\sum_{i=0}^{2^k-1} \alpha_i |i\rangle \right) \otimes |0\rangle^{\otimes k}. \quad (92)$$

Phase 3 Next, we perform k CNOT gates with qubit i (from $i = 0$ to $i = k - 1$) as controlled qubit and qubit $i + k$ as target qubit.

$$|\psi\rangle \rightarrow \sum_{i=0}^{2^k-1} \alpha_i |i\rangle |i\rangle \quad (93)$$

Phase 4 Then, apply k -qubit unitary operation U_1, U_2 on each party, respectively such that

$$U_1 |i\rangle = |i\rangle \quad U_2 |i\rangle = |i\rangle \quad (94)$$

In the case of $n = 2k - 1$, we divide the system into bipartite state by $k - 1$ qubits and k qubits.

Phase 1 From schmidt decomposition, we have:

$$|\psi\rangle = \sum_{i=0}^{2^{k-1}-1} \alpha_i |i_1\rangle |i_2\rangle. \quad (95)$$

Phase 2 Apply local unitary gate to initial state, $ket\psi = ket0^{\otimes n}$, such that

$$ket\psi \rightarrow \left(\sum_{i=0}^{2^{k-1}-1} \alpha_i |i\rangle \right) \otimes |0\rangle^{\otimes k}. \quad (96)$$

Phase 3 Apply k CNOT gates with qubit i (from $i = 0$ to $i = k - 2$) as controlled qubit and qubit $i + k$ as target qubit.

$$|\psi\rangle \rightarrow \sum_{i=0}^{2^{k-1}-1} \alpha_i |i\rangle |i\rangle \quad (97)$$

Here the last qubit state remains $|0\rangle$ at this stage, here comes the difference.

Phase 4 Then, apply $k - 1$ -qubit, k -qubit unitary operation U_1, U_2 on each party, respectively such that

$$U_1 |i\rangle = |i\rangle \quad U_2 |i\rangle = |i\rangle \quad (98)$$

The Phase 1, 2 corresponds to the state preparation. So we can perform state preparation recursively. The problem is making k -qubit unitary operation. This procedure is well introduced in the following reference[2]. To

breifly explain, there exist a representation for k -qubit unitary operation $U \in \mathbb{C}^{2^k \times 2^k}$

$$U = \begin{pmatrix} A_0 & 0 \\ 0 & A_1 \end{pmatrix} \begin{pmatrix} C & -S \\ S & C \end{pmatrix} \begin{pmatrix} B_0 & 0 \\ 0 & B_1 \end{pmatrix} \quad (99)$$

where $A_0, A_1, B_0, B_1 \in \mathbb{C}^{2^{k-1} \times 2^{k-1}}$ are unitary matrices and C, S are real diagonal matrices such that $C^2 + S^2 = 1$. This decomposition is known as Cosine-Sine Decomposition(CSD). CSD can be performed in following ways. U is partitioned into four equally sized blocks:

$$U = \begin{pmatrix} U_{00} & U_{01} \\ U_{10} & U_{11} \end{pmatrix} \quad (100)$$

Then there exist unitary matrices A_0, A_1, B_0, B_1 such that

$$U = \begin{pmatrix} A_0 & 0 \\ 0 & A_1 \end{pmatrix} \begin{pmatrix} C & -S \\ S & C \end{pmatrix} \begin{pmatrix} B_0 & 0 \\ 0 & B_1 \end{pmatrix}. \quad (101)$$

And it is actually performing four singular value decomposition.

$$U_{00} = A_0 C B_0^\dagger \quad (102)$$

$$U_{01} = A_0 S B_1^\dagger \quad (103)$$

$$U_{10} = A_1 (-S) B_0^\dagger \quad (104)$$

$$U_{11} = A_1 C B_1^\dagger \quad (105)$$

Then, it becomes controlled-U gate and controlled-Ry gate.

$$\begin{pmatrix} A_0 & 0 \\ 0 & A_1 \end{pmatrix} = |0\rangle\langle 0| \otimes A_0 + |0\rangle\langle 0| \otimes A_1 \quad (106)$$

$$\begin{pmatrix} B_0 & 0 \\ 0 & B_1 \end{pmatrix} = |0\rangle\langle 0| \otimes B_0 + |0\rangle\langle 0| \otimes B_1 \quad (107)$$

$$\begin{pmatrix} C & -S \\ S & C \end{pmatrix} = \sum I \otimes |0\rangle\langle 0| + R_y \otimes |1\rangle\langle 1| \quad (108)$$

Here, $\begin{pmatrix} A_0 & 0 \\ 0 & A_1 \end{pmatrix}$ is a controlled unitary operation, we can write it in the following form without loss of generality:

$$\begin{pmatrix} A_0 & 0 \\ 0 & A_1 \end{pmatrix} = \begin{pmatrix} V & 0 \\ 0 & V \end{pmatrix} \begin{pmatrix} D & 0 \\ 0 & D^\dagger \end{pmatrix} \begin{pmatrix} W & 0 \\ 0 & W \end{pmatrix} \quad (109)$$

where $V, W \in \mathbb{C}^{2^{k-2} \times 2^{k-2}}$ are unitary matrices, and D is a diagonal matrix. V, W, D can be found by eigen-decomposition as:

$$A_0 A_1^\dagger = V D^2 V^\dagger \quad (110)$$

$$W = D V^\dagger A_1 \quad (111)$$

In this way, the controlled unitary can be decomposed as:

$$\begin{pmatrix} A_0 & 0 \\ 0 & A_1 \end{pmatrix} = (I \otimes V)(|0\rangle\langle 0| \otimes D + |1\rangle\langle 1| \otimes D^\dagger)(I \otimes W) \quad (112)$$

Then, the unitary V, W are unitary operation in $\mathbb{C}^{2^{k-2} \times 2^{k-2}}$, which suggests that they can be iteratively decomposed by CSD. On the other hand, $(|0\rangle\langle 0| \otimes D + |1\rangle\langle 1| \otimes D^\dagger)$ is just controlled Z gate.

Problem 2 : Graph States

In some cases, one can reconstruct a quantum state using only a limited number of measurements, regardless of how many qubits are in the quantum system. One of these cases is when the quantum states are generated using only “Clifford gates” (these are essentially quantum circuits that use only H, S, CNOT and CZ gates). In this problem, we will perform quantum state tomography on a subset of Clifford states known as “graph states”. A graph state on n qubits is any state which can be generated using a circuit of the form

$$|\psi\rangle = \left(\prod_{(i,j) \in G} CZ_{i,j} \right) H^{\otimes n} |0\rangle \quad (113)$$

Where CZ_{ij} is a controlled-Z gate applied on qubits i and j , and (i,j) are edges of a hidden graph G . Different graphs, G , create different graph states. Our goal is to determine generate the graph state, $|\psi\rangle$, without knowing the specific graph which was used.

a) Look into the properties of graph states. Graph states are stabilizer states with a specific structure. Describe the key properties of stabilizer states. Using the stabilizer properties of graph states, describe how one can perform tomography on graph states.

Graph states are a class of stabilizer states with a defined structure. Stabilizer states for n -qubit systems are uniquely characterized by n stabilizers, each being an N -fold tensor product of Pauli operators (up to a factor of $\{\pm 1, \pm i\}$). A stabilizer is an operator under which the state remains invariant, making the state a simultaneous +1 eigenstate of all N stabilizer:

$$S_i |\psi\rangle = |\psi\rangle \quad (114)$$

For graph states, the stabilizer associated with each vertex i has the form:

$$S_i = X_i \prod_{(i,j) \in G} Z_j \quad (115)$$

where each stabilizer S_i corresponds to a specific qubit i and its neighbors in Graph G .

Here is an important remark. a graph state is a simultaneous $+1$ eigenstate of its stabilizers, meaning it can be represented as a linear combination of the stabilizer eigenbasis. The stabilizer eigenbasis takes the form $|\pm\rangle_i \otimes_{j \neq i} |b_j\rangle$ with condition $b_i \oplus_{j \in (i,j)} b_j = 0$, imposed by the $+1$ eigenvalue requirement (where $+, -$ correspond to $0, 1$).

To perform measurements in this basis for a given vertex i , apply a Hadamard gate H to qubit i before the measurement. After collecting multiple measurement outcomes, identify the set of j such that $b_i \oplus_{j \in (i,j)} b_j = 0$. By repeating this process for each qubit, we can reconstruct the graph of the n -qubit system by running the circuit n times.

```
def generate_clifford_circ(num_qubits, depth):
    U_hidden = qiskit.QuantumCircuit(num_qubits)
    for i in range(num_qubits):
        U_hidden.h(i)
    count = 0
    s1s2 = {-1,-1}
    while(count < depth):
        site_i = int(np.random.rand()*num_qubits)
        site_j = int(np.random.rand()*num_qubits)
        if(site_i == site_j or {site_i, site_j} == s1s2):
            continue
        print(site_i, site_j, count)
        count += 1
        s1s2 = {site_i, site_j}
        U_hidden.cz(site_i, site_j)
    return U_hidden
```

b) You may use the following code snippet to generate a random graph state

You may set `num_qubits ; 20`. Following the format of Problem 1, measure U_{hidden} in $j=20$ bases by appendix quantum circuits to U_{hidden} and taking no more than 100 shots per basis. Then create a new circuit U_{gen} such that $U_{\text{gen}}^\dagger U_{\text{hidden}} |0\rangle = |0\rangle$. As outlined in part a), I implemented the following algorithm:

Phase 1 Apply a Hadamard gate to the i -th site after generating the graph state and before measurement.

Phase 2 Extract the resulting bit strings from the measurements.

Phase 3 Identify the set of j such that $b_i \oplus j \in (i, j) \oplus b_j = 0$. Checking all possible sets of (i, j) is feasible since the total number of combinations to check is $n2^{n-1}$, which amounts to at most $20 \times 2^{19} \simeq 10485760$ per graph state.

Here are the implemented code:

```
import itertools

def GraphStateQST(U_hidden, num_qubits, num_shots=100):
    hidden_circuit = U_hidden.copy()

    graph = [] # G: set of (i,j)
    for i in range(num_qubits):
        # Phase 1
        qc = qiskit.QuantumCircuit(num_qubits)
        qc.h(i)
        circ = hidden_circuit.compose(qc)
        circ.measure_all()

        # Phase 2
        # Extract bit string
        result = backend.run(circ, shots=num_shots)
        .result().data()["counts"]
        result = [int(i,16) for i in result.keys()]

        # Phase 3
        # Generating possible set of (i,j)
        qubit_index = [idx for idx in range(num_qubits)]
        qubit_index.pop(i)

        for k in range(1, num_qubits):
            for comb in itertools.combinations(qubit_index, k):
                isConnected = xor_bits_at_positions(result, comb, i)
                if isConnected:
                    for edge in comb:
                        graph.append({i, edge}) \
                        if {i, edge} not in graph else None

    return graph
```

Here are the other functions, one to verify $b_i \oplus j \in (i, j) \oplus b_j = 0$ and

the other for generating graph state from the given graph, represented by U_{gen} .

```
# Phase 3
def xor_bits_at_positions(nums, positions, base_location):
    for num in nums:
        # Start with 0 for XOR accumulation
        result = (num >> base_location) & 1

        # XOR each specified bit position
        for pos in positions:
            # Extract the bit at the current position
            # and XOR it with the result
            result ^= (num >> pos) & 1

        if result == 0:
            continue
        else:
            return False

    return True

# Given graph G generate graph state.
def GraphState(num_qubits, graph):
    U_gen = qiskit.QuantumCircuit(num_qubits)
    for i in range(num_qubits):
        U_gen.h(i)

    for vertex in graph:
        temp = list(vertex)
        site_i = temp[0]
        site_j = temp[1]
        U_gen.cz(site_i, site_j)

    return U_gen
```

Problem 3 : Low Depth Brickwork Circuits

The N -qubit pure state $|q_{N-1}q_{N-2}\cdots q_1q_0\rangle$ requires at least $2N + 3$ additional circuit as discussed in Problem 1, Part III. $N > 10$, generally, it is impossible to perform QST exactly under the problem's constraint: limitation of the number of shots and the number of circuits. However, in some

cases, if we have primitive information about the state, we can perform a not bad QST with good approximations. In this case, we know the circuit ansatz, that is, how the state is generated. The given ‘one brickwork layer’ circuit has low circuit depth¹, 2, where each brick counts for 1. Therefore, it is assumed that the generated state has low entanglement. Especially, when the distance between q_i and q_j is far, the less entanglement is presumed.

Von Neumann entropy is one of the popular measurement of entanglement:

$$S = -\alpha_i \ln \alpha_i. \quad (116)$$

where α_i are schmidt coefficients. For a convenience, let α is sorted in a descending order. To be less entangled, we can assume, $|\alpha_0|^2 \gg |\alpha_i|^2$. In this context, I conceived three following ideas.

Idea 1. From Schmidt decomposition, we have:

$$|q_{N-1}q_{N-2} \cdots q_1q_0\rangle = \sum_{k=0}^3 \alpha_i |q_i^k q_{i+1}^k\rangle |\cdots q_{i-1}^k q_{i+2}^k \cdots\rangle \quad (117)$$

where $|q_i^k q_{i+1}^k\rangle, |\cdots q_{i-1}^k q_{i+2}^k \cdots\rangle$ are orthonormal basis for each party. Here as $|\alpha_0|^2 \gg |\alpha_k|^2$, $i \in k$, so I think I can make an approximation:

$$|q_{N-1}q_{N-2} \cdots q_1q_0\rangle \simeq |q_{i+1}^0 q_i^0\rangle |\cdots q_{i+2}^0 q_{i-1}^0 \cdots\rangle \quad (118)$$

Idea 2. From Idea 1., we see that we can approximate the ‘one Brickwork layer’ circuit as seperable state in Eq. 118. Then, we can perfrom two-qubit QST to idenify $|q_i^0 q_{i-1}^0\rangle$. For example, a POVM Ω in a certain quantum circuit should only perform measure on q_{i-1} and q_i , the estimated probability m_Ω will be given as:

$$\begin{aligned} m_\Omega &= \langle q_{N-1}q_{N-2} \cdots q_1q_0 | \Omega | q_{N-1}q_{N-2} \cdots q_1q_0 \rangle \\ &\simeq \langle q_{i+1}^0 q_i^0 | \Omega | q_{i+1}^0 q_i^0 \rangle \langle \cdots q_{i+2}^0 q_{i-1}^0 \cdots | \cdots q_{i+2}^0 q_{i-1}^0 \cdots \rangle \\ &= \langle q_{i+1}^0 q_i^0 | \Omega | q_{i+1}^0 q_i^0 \rangle \end{aligned} \quad (119)$$

In this way, we can idenify approximated state of $|q_i^0 q_{i-1}^0\rangle$.

Idea 3. In Eq. 118, we can perform one more Schmidt decomposition on the second party, with $N - 2$ qubits. Then,

$$\begin{aligned} |q_{N-1}q_{N-2} \cdots q_1q_0\rangle &\simeq \sum_{k=0}^3 \beta_k |q_{j+1}^0 q_j^0\rangle |q_{i+1}^0 q_i^0\rangle \\ &\quad |\cdots q_{j+2}^0 q_{j-1}^0 \cdots q_{i+2}^0 q_{i-1}^0 \cdots\rangle. \end{aligned} \quad (120)$$

¹the longest path in the circuit

In a similar manner, we can approximate $\beta_0 = 1$. Note that this approximation is valid when $|i - j|$ is large enough at least equal or more than 4, as the circuit depth 2. From Idea 2, we can now perform two qubit state on $q_i, q_{i+1}, q_j, q_{j+1}$ and even more, at the same time. Then, in this case, we need four detectors on $q_i, q_{i+1}, q_j, q_{j+1}$ and therefore, there are $2^4 = 16$ measurement outcomes. However, as we approximated the state into separable state, we can somewhat gather and combine measurement outcomes. For example, the measurement outcome $m_{0_i 0_{i+1}}$ can be calculated as:

$$m_{0_i 0_{i+1}} = m_{0_i 0_{i+1} 0_j 0_{j+1}} + m_{0_i 0_{i+1} 0_j 1_{j+1}} + m_{0_i 0_{i+1} 1_j 0_{j+1}} + m_{0_i 0_{i+1} 1_j 1_{j+1}} \quad (121)$$

Therefore, we can perform two-qubit QST on multiple qubit pairs at the same time with 5 circuit running. Preparing four sets of qubit pairs such that each qubit pair has distance at least 4 for each other, we can perform two-qubit QST for whole neighbors qubit pairs: $\{(q_0, q_1), (q_1, q_2), \dots (q_{N-2}, q_{N-1})\}$.

Here I tried to ...

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